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Electrostatic discharge protection device including source silicide pattern and drain silicide pattern

Abstract

An electrostatic discharge (ESD) protection device includes a semiconductor substrate, a gate structure, a source doped region, a drain doped region, source silicide patterns, and drain silicide patterns. The gate structure is disposed on the semiconductor substrate. The source doped region and the drain doped region are disposed in the semiconductor substrate and located at two opposite sides of the gate structure in a first direction, respectively. The source silicide patterns are disposed on the source doped region. The source silicide patterns are arranged in a second direction and separated from one another. The drain silicide patterns are disposed on the drain doped region. The drain silicide patterns are arranged in the second direction and separated from one another. The source silicide patterns and the drain silicide patterns are arranged misaligned with one another in the first direction.

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Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention relates to an electrostatic discharge protection device, and more particularly, to an electrostatic discharge protection device including a source silicide pattern and a

drain silicide pattern.

2. Description of the Prior Art

(2) Electrostatic discharge (ESD) is an effect in which an electric charge moves rapidly (discharge) in a short period of time through a discharge path after accumulating on a non-conductor or an ungrounded conductor. Electrostatic discharge damages the circuit formed by the components of an integrated circuit. Generally, the human body, machines used to package integrated circuits, and instruments for testing integrated circuits are all common charged bodies, and when a charged body comes in contact with a chip, the charged body may discharge to the chip. The instantaneous power of the electrostatic discharge may cause damage or failure to the integrated circuit in the chip. Therefore, ESD protection structures may be disposed in the integrated circuits for improving the problems described above.

SUMMARY OF THE INVENTION

(3) An electrostatic discharge (ESD) protection device is provided in the present invention. Source silicide patterns and drain silicide patterns arranged misaligned with one another and/or source contact structures and drain contact structures arranged misaligned with one another may be used to increase the distance between a gate structure and the silicide patterns and/or the contact structures. ESD capability and electrical over stress (EOS) capability may be enhanced accordingly, and/or an area occupied by the ESD protection device may be reduced relatively.

(4) An electrostatic discharge protection device is provided in an embodiment of the present invention. The electrostatic discharge protection device includes a semiconductor substrate, a gate structure, a source doped region, a drain doped region, source silicide patterns, and drain silicide patterns. The gate structure is disposed on the semiconductor substrate. The source doped region and the drain doped region are disposed in the semiconductor substrate and located at two opposite sides of the gate structure in a first direction, respectively. The source silicide patterns are disposed on the source doped region, and the source silicide patterns are arranged in a second direction and separated from one another. The drain silicide patterns are disposed on the drain doped region, and the drain silicide patterns are arranged in the second direction and separated from one another. The source silicide patterns and the drain silicide patterns are arranged misaligned with one another in the first direction.

(5) An electrostatic discharge protection device is provided in an embodiment of the present invention. The electrostatic discharge protection device includes a semiconductor substrate, a gate structure, a source doped region, a drain doped region, at least one source silicide pattern, at least one drain silicide pattern, first contact structures, and second contact structures. The gate structure is disposed on the semiconductor substrate. The source doped region and the drain doped region are disposed in the semiconductor substrate and located at two opposite sides of the gate structure in a first direction, respectively. The source silicide pattern is disposed on the source doped region. The drain silicide pattern is disposed on the drain doped region. The first contact structures are disposed on the source silicide pattern, and the first contact structures are arranged in a second direction and separated from one another. The second contact structures are disposed on the drain silicide pattern, and the second contact structures are arranged in the second direction and separated from one another. The first contact structures and the second contact structures are arranged misaligned with one another in the first direction.

(6) These and other objectives of the present invention will no doubt become obvious to those of ordinary skill in the art after reading the following detailed description of the preferred embodiment that is illustrated in the various figures and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic drawing illustrating an electrostatic discharge protection device according to a first embodiment of the present invention.
- (2) FIG. 2 is a schematic drawing illustrating first contact structures and second contact structures according to some embodiments of the present invention.
- (3) FIG. 3 is a schematic drawing illustrating an electrostatic discharge protection device according to the first embodiment of the present invention, which mainly shows the relative disposition relationship between source silicide patterns and drain silicide patterns.
- (4) FIG. 4 is a schematic drawing illustrating an electrostatic discharge protection device according to the first embodiment of the present invention, which mainly shows the relative disposition relationship between source contacts and drain contacts.
- (5) FIG. 5 is a schematic cross-sectional drawing illustrating a part of the electrostatic discharge protection device according to the first embodiment of the present invention.
- (6) FIG. 6 is a schematic drawing illustrating an electrostatic discharge protection device according to a second embodiment of the present invention.
- (7) FIG. 7 is a schematic drawing illustrating an electrostatic discharge protection device according to a third embodiment of the present invention.
- (8) FIG. 8 is a schematic drawing illustrating an electrostatic discharge protection device according to a fourth embodiment of the present invention.
- (9) FIG. 9 is a schematic drawing illustrating an electrostatic discharge protection device according to a fifth embodiment of the present invention.

DETAILED DESCRIPTION

- (10) The present invention has been particularly shown and described with respect to certain embodiments and specific features thereof. The embodiments set forth herein below are to be taken as illustrative rather than limiting. It should be readily apparent to those of ordinary skill in the art that various changes and modifications in form and detail may be made without departing from the spirit and scope of the present invention.
- (11) Before the further description of the preferred embodiment, the specific terms used throughout the text will be described below.
- (12) The terms “on,” “above,” and “over” used herein should be interpreted in the broadest manner such that “on” not only means “directly on” something but also includes the meaning of “on” something with an intermediate feature or a layer therebetween, and that “above” or “over” not only means the meaning of “above” or “over” something but can also include the meaning it is “above” or “over” something with no intermediate feature or layer therebetween (i.e., directly on something).
- (13) The ordinal numbers, such as “first”, “second”, etc., used in the description and the claims are used to modify the elements in the claims and do not themselves imply and represent that the claim has any previous ordinal number, do not represent the sequence of some claimed element and another claimed element, and do not represent the sequence of the manufacturing methods, unless an addition description is accompanied. The use of these ordinal numbers is only used to make a claimed element with a certain name clear from another claimed element with the same name.
- (14) The term “forming” or the term “disposing” are used hereinafter to describe the behavior of applying a layer of material to the substrate. Such terms are intended to describe any possible layer forming techniques including, but not limited to, thermal growth, sputtering, evaporation, chemical vapor deposition, epitaxial growth, electroplating, and the like.
- (15) Please refer to FIG. 1. FIG. 1 is a schematic drawing illustrating an electrostatic discharge (ESD) protection device **101** according to a first embodiment of the present invention. As shown in FIG. 1, the ESD protection device **101** includes a semiconductor substrate **10**, at least one gate structure **20**, at least one source doped region **30S**, at least one drain doped region **30D**, at least one

source silicide pattern **50S**, at least one drain silicide pattern **50D**, at least one first contact structure **CT1**, and at least one second contact structure **CT2**. The gate structure **20** is disposed on the semiconductor substrate **10**. The source doped region **30S** and the drain doped region **30D** are disposed in the semiconductor substrate **10** and located at two opposite sides of the gate structure **20** in a first direction (such as a direction **D1** shown in FIG. **1**), respectively. The source silicide pattern **50S** is disposed on the source doped region **30S**, and the drain silicide pattern **50D** is disposed on the drain doped region **30D**. The first contact structure **CT1** is disposed on the source silicide pattern **50S**, and the second contact structure **CT2** is disposed on the drain silicide pattern **50D**.

(16) In some embodiments, the ESD protection device **101** may include a plurality of the source silicide patterns **50S** and a plurality of the drain silicide patterns **50D**, but not limited thereto. The source silicide patterns **50S** may be disposed on the source doped region **30S** in a vertical direction (such as a direction **D3** shown in FIG. **1**), the drain silicide patterns **50D** may be disposed on the drain doped region **30D** in the direction **D3**, and the direction **D3** may be regarded as a thickness direction of the semiconductor substrate **10**. In addition, the source silicide patterns **50S** may be arranged in a second direction (such as a direction **D2** shown in FIG. **1**) and separated from one another, and the drain silicide patterns **50D** may be arranged in the direction **D2** and separated from one another also. The source silicide patterns **50S** and the drain silicide patterns **50D** may be arranged misaligned with one another in the direction **D1**. In other words, each of the source silicide patterns **50S** is not aligned with any one of the drain silicide patterns **50D** in the direction **D1**, and each of the drain silicide patterns **50D** is not aligned with any one of the source silicide patterns **50S** in the direction **D1**.

(17) In some embodiments, the ESD protection device **101** may include a plurality of the first contact structures **CT1** and a plurality of the second contact structures **CT2**, but not limited thereto. The first contact structures **CT1** may be disposed on the source silicide patterns **50S**, respectively, and the second contact structures **CT2** may be disposed on the drain silicide patterns **50D**, respectively, but not limited thereto. In addition, the first contact structures **CT1** may be arranged in the direction **D2** and separated from one another, and the second contact structures **CT2** may be arranged in the direction **D2** and separated from one another. In some embodiments, each of the first contact structures **CT1** may be disposed on one of the source silicide patterns **50S** in the direction **D3**, and each of the second contact structures **CT2** may be disposed on one of the drain silicide patterns **50D** in the direction **D3**. In other words, there may be only one first contact structure **CT1** disposed on each source silicide pattern **50S**, and there may be only one second contact structure **CT2** disposed on each drain silicide pattern **50D**, but not limited thereto. Additionally, the first contact structures **CT1** and the second contact structures **CT2** may be arranged misaligned with one another in the direction **D1**. In other words, each of the first contact structures **CT1** is not aligned with any one of the second contact structures **CT2** in the direction **D1**, and each of the second contact structures **CT2** is not aligned with any one of the first contact structures **CT1** in the direction **D1**.

(18) Specifically, in some embodiments, the direction **D3** described above may be regarded as a thickness direction of the semiconductor substrate **10**, the semiconductor substrate **10** may have a top surface and a bottom surface opposite to the top surface in the direction **D3**, and the gate structure **20**, the first contact structures **CT1**, and the second contact structures **CT2** described above may be disposed at a side of the top surface. Additionally, in some embodiments, the direction **D1** may be substantially orthogonal to the direction **D2**, and the direction **D1** and the direction **D2** may be substantially orthogonal to the direction **D3** respectively. Therefore, the direction **D1** and the direction **D2** may be regarded as horizontal directions parallel with the top surface and/or the bottom surface of the semiconductor substrate **10**, but not limited thereto. In this description, a distance between the bottom surface of the semiconductor substrate **10** and a relatively higher location and/or a relatively higher part in the direction **D3** is greater than a

distance between the bottom surface of the semiconductor substrate **10** and a relatively lower location and/or a relatively lower part in the direction **D3**. The bottom or a lower portion of each component may be closer to the bottom surface of the semiconductor substrate **10** in the direction **D3** than the top or upper portion of this component. Another component disposed above a specific component may be regarded as being relatively far from the bottom surface of the semiconductor substrate **10** in the direction **D3**, and another component disposed under a specific component may be regarded as being relatively closer to the bottom surface of the semiconductor substrate **10** in the direction **D3**.

(19) In some embodiments, two of the source silicide patterns **50S** located adjacent to each other in the direction **D2** may be separated from each other by a gap **GP1**, and two of the drain silicide patterns **50D** located adjacent to each other in the direction **D2** may be separated from each other by a gap **GP2**. In other words, there is not any silicide pattern disposed in the gap **GP1**, and there is not any silicide pattern disposed in the gap **GP2** also. In addition, because the source silicide patterns **50S** and the drain silicide patterns **50D** are arranged misaligned with one another in the direction **D1**, one of the drain silicide patterns **50D** may overlap the gap **GP1** when viewed in the direction **D1** and/or be aligned with the gap **GP1** in the direction **D1**, and one of the source silicide patterns **50S** may overlap the gap **GP2** when viewed in the direction **D1** and/or be aligned with the gap **GP2** in the direction **D1**. It is worth noting that, in this description, when two objects overlap in a specific direction, it can be considered that when the two objects are viewed in the specific direction, at least some regions of the two objects overlap each other. In some embodiments, in a top view diagram of the ESD protection device **101** (such as FIG. 1), a length of the gap **GP1** in the direction **D2** may be substantially equal to a length of the gap **GP2** in the direction **D2**. For instance, the length of the gap **GP1** in the direction **D2** may be substantially equal to the length of the gap **GP2** in the direction **D2** with a tolerance of $\pm 10\%$ or a tolerance of $\pm 5\%$. An area of each of the source silicide patterns **50S** may be substantially equal to an area of each of the drain silicide patterns **50D**. For instance, the area of each of the source silicide patterns **50S** in the direction **D3** may be substantially equal to the area of each of the drain silicide patterns **50D** in the direction **D3** with a tolerance of $\pm 10\%$ or a tolerance of $\pm 5\%$, but not limited thereto. In other words, the dimension of each of the source silicide patterns **50S** may be substantially the same as that of each of the drain silicide patterns **50D**, and the arrangement pitch of the source silicide patterns **50S** in the direction **D2** may be substantially equal to that of the drain silicide patterns **50D** in the direction **D2**, but not limited thereto.

(20) In some embodiments, each of the first contact structures **CT1** may include one or a plurality of source contacts **60S** disposed on the corresponding source silicide patterns **50S**, and the source contacts **60S** may be separated from one another without being connected with one another. Each of the second contact structures **CT2** may include one or a plurality of drain contacts **60D** disposed on the corresponding drain silicide patterns **50D**, and the drain contacts **60D** may be separated from one another without being connected with one another. In some embodiments, each of the first contact structures **CT1** may consist of one source contact **60S** or a plurality of the source contacts **60S**, and each of the second contact structures **CT2** may consist of one drain contact **60D** or a plurality of the drain contacts **60D**, but not limited thereto. For example, in some embodiments, each of the first contact structures **CT1** may include four source contact **60S** arranged in an array configuration and separated from one another, and each of the second contact structures **CT2** may include four drain contact **60D** arranged in an array configuration and separated from one another, but not limited thereto.

(21) Please refer to FIG. 1 and FIG. 2. FIG. 2 is a schematic drawing illustrating the first contact structures and the second contact structures according to some embodiments of the present invention. As shown in FIG. 1 and FIG. 2, in some embodiments, each of the first contact structures **CT1** may include one or a plurality of source contacts **60S**, and each of the second contact structures **CT2** may include one or a plurality of drain contacts **60D**. The source contacts

60S in each of the first contact structures **CT1** may be arranged and disposed uniformly on the corresponding source silicide pattern **50S** with substantially the same spacing therebetween, and the drain contacts **60D** in each of the second contact structures **CT2** may be arranged and disposed uniformly on the corresponding drain silicide pattern **50D** with substantially the same spacing therebetween, but not limited thereto. Additionally, the dimensions and/or the shapes of the source silicide patterns **50S** and the drain silicide patterns **50D** may be adjusted in accordance with the numbers and/or the arrangements of the source contacts **60S** in the first contact structure **CT1** and the drain contacts **60D** in the second contact structure **CT2**.

(22) As shown in FIG. 1, in some embodiments, two of the first contact structures **CT1** located adjacent to each other in the direction **D2** may be separated from each other by a gap **GP3**, and two of the second contact structures **CT2** located adjacent to each other in the direction **D2** may be separated from each other by a gap **GP4**. In other words, there is not any contact structure and/or contact disposed in the gap **GP3**, and there is not any contact structure or contact disposed in the gap **GP4** also. In addition, because the first contact structures **CT1** and the second contact structures **CT2** are arranged misaligned with one another in the direction **D1**, one of the second contact structures **CT2** may overlap the gap **GP3** when viewed in the direction **D1** and/or be aligned with the gap **GP3** in the direction **D1**, and one of the first contact structures **CT1** may overlap the gap **GP4** when viewed in the direction **D1** and/or be aligned with the gap **GP4** in the direction **D1**. In some embodiments, a distance **DS1** between two of the source contacts **60S** disposed in each first contact structure **CT1** and located adjacent to each other in the direction **D2** may be less than a length of the gap **GP3** in the direction **D2**, and the length of the gap **GP1** in the direction **D2** may be less than the length of the gate **GP3** in the direction **D2**. A distance **DS2** between two of the drain contacts **60D** disposed in each second contact structure **CT2** and located adjacent to each other in the direction **D2** may be less than a length of the gap **GP4** in the direction **D2**, and the length of the gap **GP2** in the direction **D2** may be less than the length of the gate **GP4** in the direction **D2**.

(23) In some embodiments, one of the drain silicide patterns **50D** may overlap the gap **GP1** and two of the source silicide patterns **50S** located adjacent to each other and connected with the gap **GP1** when viewed in the direction **D1**, one of the source silicide patterns **50S** may overlap the gap **GP2** and two of the drain silicide patterns **50D** located adjacent to each other and connected with the gap **GP2** when viewed in the direction **D1**, and each of the first contact structures **CT1** and each of the second contact structures **CT2** may not overlap when viewed in the direction **D1**, but not limited thereto. Additionally, in some embodiments, a distance between the gate structure **20** and one of the source silicide patterns **50S** in the direction **D1** (such as a distance **DS11** illustrated in FIG. 1) may be substantially equal to a distance between the gate structure **20** and one of the drain silicide patterns **50D** in the direction **D1** (such as a distance **DS12** illustrated in FIG. 1). For instance, the distance **DS11** may be substantially equal to the distance **DS12** with a tolerance of $\pm 10\%$ or a tolerance of $+5\%$, but not limited thereto. However, the relative relationship between the distance **DS11** and the distance **DS12** may be adjusted in accordance with different product specifications, and the distance **DS11** may be different from the distance **DS12** in some embodiments accordingly.

(24) Please refer to FIG. 1 and FIG. 3. FIG. 3 is a schematic drawing illustrating the ESD protection device according to the first embodiment of the present invention, which mainly shows the relative disposition relationship between the source silicide patterns and the drain silicide patterns, and the first contact structures **CT1** and the second contact structures **CT2** described above are not illustrated in FIG. 3. As shown in FIG. 1 and FIG. 3, in the top view diagram of the ESD protection device **101**, a center point **CP1** of one of the source silicide patterns **50S** and a center point **CP2** of one of the drain silicide patterns **50D** may be located on a straight line **L1** extending in a first oblique direction (such as a direction **D4** shown in FIG. 3), the direction **D4** is not parallel with the direction **D1**, and the direction **D4** is not parallel with the direction **D2**. For

example, an included angle A1 between the direction D4 and the direction D1 may be greater than 0 degree and less than 90 degrees, but not limited thereto. In addition, the center point CP1 and the center point CP2 described above may be a geometric center of the source silicide pattern 50S and a geometric center of the drain silicide pattern 50D, respectively, in the top view diagram of the ESD protection device.

(25) Please refer to FIG. 4. FIG. 4 is a schematic drawing illustrating the ESD protection device according to the first embodiment of the present invention, which mainly shows the relative disposition relationship between source contacts and drain contacts. As shown in FIG. 4, in the top view diagram of the ESD protection device 101, a length of a straight line L2 connected with one of the source contacts 60S and one of the drain contacts 60D may be equal to the shortest distance between the source contact 60S and the drain contact 60D (such as a distance DS3 illustrated in FIG. 4), and the straight line L2 may extend in a second oblique direction (such as a direction D5 shown in FIG. 4). The direction D5 is not parallel with the direction D1, and the direction D5 is not parallel with the direction D2. For example, an included angle A2 between the direction D5 and the direction D1 may be greater than 0 degree and less than 90 degrees, but not limited thereto.

(26) Please refer to FIG. 1, FIG. 4, and FIG. 5. FIG. 5 is a schematic cross-sectional drawing illustrating a part of the ESD protection device according to the first embodiment of the present invention, such as a cross-sectional diagram taken along a line parallel with the direction D5, but not limited thereto. As shown in FIG. 1, FIG. 4, and FIG. 5, with the source silicide patterns 50S and the drain silicide patterns 50D arranged misaligned with one another and/or with the first contact structures CT1 and the second contact structures CT2 arranged misaligned with one another, the shortest distance between the source contact 60S and the drain contact 60D (such as the distance DS3 illustrated in FIG. 4) may be increased. Accordingly, a distance DS21 between the source contact 60S and the gate structure 20 in the direction D5 and a distance DS22 between the drain contact 60D and the gate structure 20 in the direction D5 may be increased, and a resistance of a ballast resistor formed with the material between the source contact 60S and the gate structure 20 (such as the source doped region 30S and/or the source silicide pattern 50S) and/or a resistance of a ballast resistor formed with the material between the drain contact 60D and the gate structure 20 (such as the drain doped region 30D and/or the drain silicide pattern 50D) may be increased for improving the electrical performance of the ESD protection device.

(27) Specifically, in some embodiments, the semiconductor substrate 10 may include a silicon substrate, a silicon germanium substrate, or a semiconductor substrate formed with other suitable materials, and the ESD protection device 101 may further include a well region 10W disposed in the semiconductor substrate 10. The well region 10W may be a doped well region with a first conductivity type, and the source doped region 30S and the drain doped region 30D may be doped regions with a second conductivity type complementary to the first conductivity type. Therefore, a parasite bipolar junction transistor (BJT) may be formed with the source doped region 30S, the well region 10W, and the drain doped region 30D. For instance, the well region 10W may be a p-type well region, and the source doped region 30S and the drain doped region 30D may be n-type doped regions, but not limited thereto. The resistance of the ballast resistor described above may be increased by increasing the distance DS21 and/or the distance DS22 described above, and the current path in the bipolar junction transistor may be moved downwards accordingly for improving the electrostatic discharge (ESD) capability and electrical over stress (EOS) capability.

(28) In some embodiments, the ESD protection device 101 may further include a spacer structure 26 disposed on a sidewall of the gate structure 20, and the gate structure 20 may include a gate dielectric layer 22 and a gate electrode 24 disposed on the gate dielectric layer 22. The spacer structure 26 may include a single layer or multiple layers of insulation materials, such as silicon oxide, silicon nitride, or other suitable insulation materials. The gate dielectric layer 22 may include high dielectric constant (high-k) dielectric material or other suitable dielectric materials, and the gate electrode 24 may include a non-metallic electrically conductive material (such as

doped polysilicon) or a metallic electrically conductive material, but not limited thereto. In addition, the source silicide pattern **50S** and the drain silicide pattern **50D** may include cobalt-silicide, nickel-silicide, or other suitable electrically conductive metal silicide materials. The source contact **60S** and the drain contact **60D** may include a barrier layer (not illustrated) and a low resistivity material (not illustrated) disposed on the barrier layer, but not limited thereto. The low resistivity material described above may include a material with relatively lower electrical resistivity, such as copper, aluminum, tungsten, and so forth, and the barrier layer described above may include titanium nitride, tantalum nitride, or other suitable electrically conductive barrier materials, but not limited thereto.

(29) As shown in FIG. 1 and FIG. 5, each of the source silicide patterns **50S** is separated from the gate structure **20** and the spacer structure **26**, and each of the drain silicide patterns **50D** is separated from the gate structure **20** and the spacer structure **26**. In some embodiments, the ESD protection device **101** may further include a silicide block pattern **40** disposed on the semiconductor substrate **10**, the source silicide patterns **50S** may be separated from one another by the silicide block pattern **40**, and the drain silicide patterns **50D** may be separated from one another by the silicide block pattern **40**. In some embodiments, in the top view diagram of the ESD protection device, each of the source silicide patterns **50S** and each of the drain silicide patterns **50D** may be surrounded by the silicide block pattern **40**. The locations and shapes of each source silicide pattern **50S** and each drain silicide pattern **50D** may be defined by the silicide block pattern **40**. In some embodiments, the silicide block pattern **40** may be formed with an insulation hard mask. In some embodiments, the silicide block pattern **40** may be formed with a photoresist material and may be removed after the source silicide patterns **50S** and the drain silicide patterns **50D** are formed.

(30) As shown in FIG. 1, in some embodiments, the ESD protection device **101** may include a plurality of the gate structures **20** and a plurality of the source doped regions **30S**, and the ESD protection device **101** may further include an isolation structure **12**, a doped region **14**, contacts **60P**, and gate contacts **60G**. For example, the ESD protection device **101** may include two gate structures **20** disposed on the semiconductor substrate **10**, and each of the gate structures **20** may be elongated in the direction **D2**. The drain doped region **30D**, the drain silicide patterns **50D**, and the second contact structures **CT2** may be disposed between the two gate structures **20** in the direction **D1**. The two source doped regions **30S**, the corresponding source silicide patterns **50S** and the corresponding first contact structures **CT1** may be disposed at the relatively outer sides of the two gate structures **20** in the direction **D1**, respectively.

(31) In some embodiments, the doped region **14** may be a doped region with the first conductivity type, such as a p-type doped region, but not limited thereto. The isolation structure **12** may include a field oxide layer, a trench isolation structure, or other suitable insulating isolation structures. The isolation structure **12** may surround the source doped regions **30S** and the drain doped region **30D**, and the isolation structure **12** may be disposed between the doped region **14** and the source doped region **30S** and disposed between the doped region **14** and the drain doped region **30D**. The gate contacts **60G** may be disposed on the gate structures **20** for being electrically connected with the gate structures **20**, and the contacts **60P** may be disposed on the doped region **14**. In some embodiments, the material composition of the gate contacts **60G** and the contacts **60P** may be similar to that of the source contacts **60S** and the drain contacts **60D** described above, but not limited thereto. It is worth noting that the layout design of the ESD protection device **101** in the present invention is not limited to the layout design illustrated in FIG. 1 and may be further modified according to some design considerations.

(32) The following description will detail the different embodiments of the present invention. To simplify the description, identical components in each of the following embodiments are marked with identical symbols. For making it easier to understand the differences between the embodiments, the following description will detail the dissimilarities among different embodiments

and the identical features will not be redundantly described.

(33) Please refer to FIG. 6. FIG. 6 is a schematic drawing illustrating an ESD protection device **102** according to a second embodiment of the present invention. As shown in FIG. 6, in the ESD protection device **102**, there may be only one first contact structure CT1 and only one second contact structure CT2 disposed on the two opposite sides of the gate structure **20** in the direction D1, respectively, and the first contact structure CT1 and the second contact structure CT2 may be disposed corresponding to each other, overlap, and/or aligned with each other when viewed in the direction D1. In other words, in this embodiment, the first contact structure CT1 and the second contact structure CT2 are not arranged misaligned with each other in the direction D1.

Comparatively, the source silicide pattern **50S** and the drain silicide pattern **50D** in this embodiment may be disposed corresponding to each other, overlap, and/or aligned with each other when viewed in the direction D1. Therefore, the source silicide pattern **50S** and the drain silicide pattern **50D** are not arranged misaligned with each other in the direction D1, also. In this embodiment, a length of a straight line L3 connected with one of the source contacts **60S** and one of the drain contacts **60D** may be equal to the shortest distance between the source contact **60S** and the drain contact **60D** (such as a distance DS4 illustrated in FIG. 6), and the straight line L3 extends in the direction D1.

(34) Please refer to FIG. 6 and FIG. 4. As shown in FIG. 6 and FIG. 4, when the area occupied by the ESD protection device **102** in the direction D3 is substantially equal to the area occupied by the ESD protection device **101** in the direction D3, the shortest distance between the source contact **60S** and the drain contact **60D** in the ESD protection device **101** (such as the distance DS3) may be greater than the shortest distance between the source contact **60S** and the drain contact **60D** in the ESD protection device **102** (such as the distance DS4). Therefore, the ESD protection device **101** may have better electrical performance (such as better ESD capability and better EOS capability). Relatively, when the distance DS3 is substantially equal to the distance DS4, the related electrical performance (such as the ESD capability and the EOS capability) of the ESD protection device **101** may be similar to that of the ESD protection device **102**, but the area occupied by the ESD protection device **101** may be relatively smaller (the length in the direction D1 may be reduced, for example) because of the source silicide patterns **50S** and the drain silicide patterns **50D** arranged misaligned with one another and/or the first contact structures CT1 and the second contact structures CT2 arranged misaligned with one another, and the purpose of reducing the occupied area may be achieved accordingly.

(35) Please refer to FIG. 7. FIG. 7 is a schematic drawing illustrating an ESD protection device **103** according to a third embodiment of the present invention. As shown in FIG. 7, in the ESD protection device **103**, the amount of the source silicide patterns **50S** and the amount of the drain silicide patterns **50D** located at two opposite sides of the same gate structure **20** in the direction D1 may be different from each other, and the amount of the first contact structures CT1 and the amount of the second contact structures CT2 located at two opposite sides of the same gate structure **20** in the direction D1 may be different from each other. For example, as shown in FIG. 7, the amount of the drain silicide patterns **50D** may be less than the amount of the source silicide patterns **50S** located at another side of the same gate structure **20** in the direction D1, and the amount of the corresponding second contact structures CT2 may be less than the amount of the first contact structures CT1. Therefore, the integrated length of the source silicide patterns **50S** in the direction D2 may be greater than the integrated length of the drain silicide patterns **50D** in the direction D2, and the integrated length of the first contact structures CT1 in the direction D2 may be greater than the integrated length of the second contact structures CT2 in the direction D2. Comparatively, in some embodiments (such as the ESD protection device **101** in FIG. 1 described above), the amount of the source silicide patterns **50S** may be less than the amount of the drain silicide patterns **50D** located at another side of the same gate structure **20** in the direction D1, and the amount of the corresponding first contact structures CT1 may be less than the amount of the second contact

structures CT2.

(36) Please refer to FIG. 8. FIG. 8 is a schematic drawing illustrating an ESD protection device **104** according to a fourth embodiment of the present invention. As shown in FIG. 8, the ESD protection device **104** may include only one drain silicide pattern **50D** disposed between the two gate structures **20** in the direction **D1**, and the second contact structures CT2 may be disposed on the same drain silicide pattern **50D**. In this embodiment, the length of the drain silicide pattern **50D** in the direction **D2** is greater than the length of each source silicide pattern **50S** in the direction **D2** and the integrated length of the source silicide patterns **50S** in the direction **D2**, but a certain degree of the ESD capability and the EOS capability improvement can still be achieved and/or the area occupied by the ESD protection device may be reduced by the first contact structures CT1 and the second contact structures CT2 arranged misaligned with one another.

(37) Please refer to FIG. 9. FIG. 9 is a schematic drawing illustrating an ESD protection device **105** according to a fifth embodiment of the present invention. As shown in FIG. 9, in the ESD protection device **105**, there may be only one source silicide pattern **50S** disposed on each source doped region **30S**, and the first contact structures CT1 may be disposed on the same source silicide pattern **50S**. In this embodiment, the length of each source silicide pattern **50S** in the direction **D2** is greater than the length of each drain silicide pattern **50D** in the direction **D2**, and the source silicide pattern **50S** may overlap the two gaps GP2 and the drain silicide pattern **50D** located between the two gaps GP2, but a certain degree of the ESD capability and the EOS capability improvement can still be achieved and/or the area occupied by the ESD protection device may be reduced by the first contact structures CT1 and the second contact structures CT2 arranged misaligned with one another.

(38) To summarize the above descriptions, according to the ESD protection device in the present invention, the source silicide patterns and the drain silicide patterns arranged misaligned with one another and/or the first contact structures and the second contact structures arranged misaligned with one another may be used to increase the distance between the gate structure and the first contact structure and/or the distance between the gate structure and the second contact structure. The ESD capability and the EOS capability may be enhanced accordingly, and/or the area occupied by the ESD protection device may be reduced relatively.

(39) Those skilled in the art will readily observe that numerous modifications and alterations of the device and method may be made while retaining the teachings of the invention. Accordingly, the above disclosure should be construed as limited only by the metes and bounds of the appended claims.

Claims

1. An electrostatic discharge (ESD) protection device, comprising: a semiconductor substrate; a first gate structure and a second gate structure, wherein the first gate structure and the second gate structure are disposed on the semiconductor substrate; a first source doped region, a second source doped region, and a drain doped region, wherein the first source doped region, the second source doped region, and the drain doped region are disposed in the semiconductor substrate, the first source doped region and the drain doped region are located at two opposite sides of the first gate structure in a first direction, respectively, and the second source doped region and the drain doped region are located at two opposite sides of the second gate structure in the first direction, respectively; first source silicide patterns disposed on the first source doped region, wherein the first source silicide patterns are arranged in a second direction and separated from one another; second source silicide patterns disposed on the second source doped region, wherein the second source silicide patterns are arranged in the second direction and separated from one another; drain silicide patterns disposed on the drain doped region, wherein the drain silicide patterns are arranged in the second direction and separated from one another, the first source silicide patterns and the

drain silicide patterns are arranged misaligned with one another in the first direction, and the second source silicide patterns and the drain silicide patterns are arranged misaligned with one another in the first direction; a silicide block pattern disposed on the semiconductor substrate, wherein the first source silicide patterns are separated from one another by the silicide block pattern, the drain silicide patterns are separated from one another by the silicide block pattern, and in a top view diagram of the ESD protection device, each of the first source silicide patterns and each of the drain silicide patterns are surrounded by the silicide block pattern; and first source contacts disposed on the first source silicide patterns and separated from one another, wherein at least four of the first source contacts are disposed on and directly contact one of the first source silicide patterns, an amount of the drain silicide patterns surrounded by the silicide block pattern is greater than an amount of the first source silicide patterns surrounded by the silicide block pattern, and the at least four first source contacts are arranged in an array configuration with multiple columns and multiple rows in the top view diagram of the ESD protection device, wherein two of the first source silicide patterns located adjacent to each other are separated by a first gap, and one of the drain silicide patterns overlaps the two of the first source silicide patterns located adjacent to each other and the first gap when viewed in the first direction, wherein two of the second source silicide patterns located adjacent to each other are separated by a second gap, and the one of the drain silicide patterns overlaps the two of the second source silicide patterns located adjacent to each other and the second gap when viewed in the first direction.

2. The ESD protection device according to claim 1, wherein two of the drain silicide patterns located adjacent to each other are separated by a third gap, and one of the first source silicide patterns overlaps the third gap when viewed in the first direction.

3. The ESD protection device according to claim 1, further comprising: a spacer structure disposed on a sidewall of the first gate structure, wherein the first source silicide patterns are separated from the first gate structure and the spacer structure.

4. The ESD protection device according to claim 1, wherein the silicide block pattern is formed with an insulation hard mask.

5. The ESD protection device according to claim 1, wherein in the top view diagram of the ESD protection device, a center point of one of the first source silicide patterns and a center point of one of the drain silicide patterns are located on a straight line extending in an oblique direction, the oblique direction is not parallel with the first direction, and the oblique direction is not parallel with the second direction.

6. The ESD protection device according to claim 1, further comprising: drain contacts disposed on the drain silicide patterns, wherein in the top view diagram of the ESD protection device, a length of a straight line connected with one of the first source contacts and one of the drain contacts is equal to the shortest distance between the first source contacts and the drain contacts, and the straight line extends in an oblique direction, wherein the oblique direction is not parallel with the first direction, and the second oblique direction is not parallel with the second direction.

7. The ESD protection device according to claim 1, wherein the first direction is orthogonal to the second direction.

8. The ESD protection device according to claim 1, wherein a distance between the first gate structure and one of the first source silicide patterns in the first direction is different from a distance between the first gate structure and one of the drain silicide patterns in the first direction.

9. The ESD protection device according to claim 1, wherein a distance between the first gate structure and one of the first source silicide patterns in the first direction is substantially equal to a distance between the first gate structure and one of the drain silicide patterns in the first direction.

10. An electrostatic discharge (ESD) protection device, comprising: a semiconductor substrate; a gate structure disposed on the semiconductor substrate; a source doped region and a drain doped region disposed in the semiconductor substrate and located at two opposite sides of the gate structure in a first direction, respectively; source silicide patterns disposed on the source doped

region and separated from one another; drain silicide patterns disposed on the drain doped region and separated from one another; a silicide block pattern disposed on the semiconductor substrate, wherein the source silicide patterns are separated from one another by the silicide block pattern, the drain silicide patterns are separated from one another by the silicide block pattern, and in a top view diagram of the ESD protection device, each of the source silicide patterns and each of the drain silicide patterns are surrounded by the silicide block pattern; first contact structures disposed on the source silicide patterns, wherein the first contact structures are arranged in a second direction and separated from one another, and each of the first contact structures is disposed on one of the source silicide patterns and comprises at least four source contacts separated from one another and directly contacting the one of the source silicide patterns; and second contact structures disposed on the drain silicide patterns, wherein the second contact structures are arranged in the second direction and separated from one another, each of the second contact structures is disposed on one of the drain silicide patterns, and the first contact structures and the second contact structures are arranged misaligned with one another in the first direction, wherein an amount of the drain silicide patterns surrounded by the silicide block pattern is greater than an amount of the source silicide patterns surrounded by the silicide block pattern, and the at least four source contacts of each of the first contact structures are arranged in an array configuration with multiple columns and multiple rows in the top view diagram of the ESD protection device, wherein two of the source silicide patterns located adjacent to each other are separated by a first gap, and one of the drain silicide patterns overlaps the two of the source silicide patterns located adjacent to each other and the first gap when viewed in the first direction.

11. The ESD protection device according to claim 10, wherein two of the first contact structures located adjacent to each other are separated by a second gap, and one of the second contact structures overlaps the second gap when viewed in the first direction.

12. The ESD protection device according to claim 11, wherein a distance between the source contacts in the second direction is less than a length of the second gap in the second direction.

13. The ESD protection device according to claim 10, wherein two of the second contact structures located adjacent to each other are separated by a second gap, and one of the first contact structures overlaps the second gap when viewed in the first direction.

14. The ESD protection device according to claim 13, wherein each of the second contact structures comprises drain contacts separated from one another, and a distance between the drain contacts in the second direction is less than a length of the second gap in the second direction.
